Question 9

[Maximum mark: 7]

Clive enjoys baking cookies. His original recipe is popular, but he has a new recipe, and is interested to see whether people prefer it to his original recipe. He bakes 50 of each type of cookie, gives one of each to 50 people, and asks them to say which one they prefer. He does not tell them which cookie is made from either recipe.

After receiving the preferences of the 50 people, Clive conducts a test at the 5% significance level, with the following null and alternative hypotheses:

$$H_0 : p = 0.5, \qquad H_1 : p > 0.5.$$

(a) Show that the critical value for Clive's test is 32. [3]

Answer:

$(a) H_0 : p = 0.5, \qquad H_1 : p > 0.5$

Let X be the number of 50 people who prefers the new recipe. So that under H_0

$$X \sim B(50, 0.5)$$

- fixed no. of trials → n=50
- Prob. is constant 0.5
- only two possible outcomes
- Indp. trials

Now $P(31 \leq x \leq 50) = 0.05946022$

and $P(x \geq 32) = 0.03245432$

$P(32 \leq x \leq 50) = 0.03245...$

Since $P(X \geq 31) > 0.05$

but $P(x \geq 32) < 0.05$

Therefore Critical value is 32

(b) State the probability that Clive makes a type I error in his test conclusion. [1]

(c) Given that 55% of people prefer the new recipe, calculate the probability of a type II error. [3]

Answer:

(b) Type I error occurs when H_0 is rejected, given that H_0 is true, so

$$P(typeIerror) = P(x \geq 32 | p = 0.5)$$

So $P(typeIerror) = 0.0325$

(c) Type II error occurs when H_0 is not rejected, given that H_0 is false, so given that p = 0.55

$$P(TypeIIerror) = P(acceptH_0 | p = 0.55)$$
$$= P(X \leq 31 | P = 0.55)$$

Using GDC $P(TypeIIerrorr) = 0.8726...$

$P(TypeIIerrorr) = 0.873$

Question 10

[Maximum mark: 8]

Two lines L_1 and L_2 are given by

$$L_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} p+2 \\ 2 \\ -p \end{pmatrix}, \qquad L_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -5 \\ p-3 \end{pmatrix},$$

where $p \in \mathbb{R}$.

(a) When $p = 1$ find the acute angle between the lines. [3]

(b) Find the values of p for which L_1 and L_2 are perpendicular. [3]

(c) Verify that L_1 and L_2 have no point of intersection if they are perpendicular. [2]

Answer:

(a) p=1, so direction vectors of lines are

$$\vec{d_1} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}, \qquad \vec{d}_2 = \begin{pmatrix} 2 \\ -5 \\ -2 \end{pmatrix}$$

$$\therefore \vec{d}_1 \cdot \vec{d}_2 = (3)(2) + (2)(-5) + (-1)(-2) = -2$$

$$\left|\vec{d}_1\right| = \sqrt{9+4+1} = \sqrt{14}, \quad \left|\vec{d}_2\right| = \sqrt{4+25+4} = \sqrt{33}$$

$$\therefore Q = \cos^{-1}\left(\frac{\left|\vec{d}_1 \cdot \vec{d}_2\right|}{\left|\vec{d}_1\right|\left|\vec{d}_2\right|}\right) = \cos^{-1}\left(\frac{|-2|}{(\sqrt{14})(\sqrt{33})}\right) \approx 84.7^\circ$$

$\cos\theta = \dfrac{|\vec{d_1} \cdot \vec{d_2}|}{|\vec{d_1}|\,|\vec{d_2}|}$

$\vec{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \vec{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$

$\vec{a} \cdot \vec{b} = (a_1)(b_1) + (a_2)(b_2) + (a_3)(b_3)$

(b) If $L_1 \perp L_2 \Rightarrow \vec{d_1} \cdot \vec{d_2} = 0$

$$\therefore \begin{pmatrix} p+2 \\ 2 \\ -p \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -5 \\ p-3 \end{pmatrix} = 0$$

$$(p+2)(2) + (2)(-5) + (-p)(p-3) = 0$$

$$2p + 4 - 10 - p^2 + 3p = 0$$

$$p^2 - 5p + 6 = 0$$

$$(p-2)(p-3) = 0$$

$$p = 2, p = 3$$

(c) Verify that L_1 and L_2 have no point of intersection if they are perpendicular. [2]

Answer:

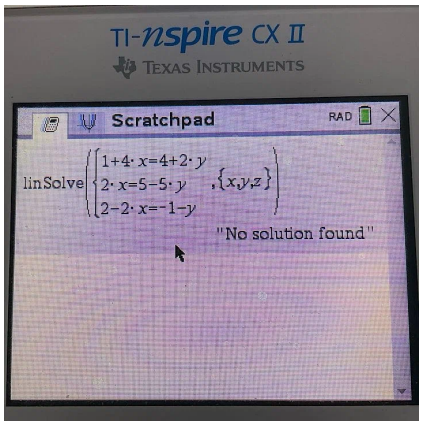
If they are perpendicular, P=2

$$L_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}, \quad L_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}$$

If the lines intersect, then there must be a solution to the system of equations

$$\begin{aligned} 1 + 4\lambda &= 4 + 2\mu \\ 2\lambda &= 5 - 5\mu \\ 2 - 2\lambda &= -1 - \mu \end{aligned}$$

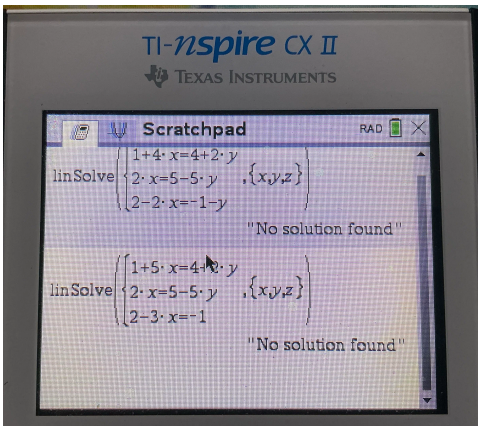
Using G.D.C., we find that there are no solution.



Similarly, in the case where p = 3, we have

$$L_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ -3 \end{pmatrix}, \quad L_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -5 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 1 + 5\lambda &= 4 + 2\mu \\ 2\lambda &= 5 - 5\mu \\ 2 - 3\lambda &= -1 \end{aligned}$$



Again, using G.D.C., we find that there are no solution.

Therefore, we have verified that:

L_1 and L_2 have no point of intersection if they are perpendicular.

Question 11

[Maximum mark: 7]

The number of emails arriving at a company help desk is modelled by a Poisson distribution with a mean of 1.2 emails per minute.

(a) Under this model, calculate the probability that

(i) at least 2 emails arrive in a particular minute;

(ii) exactly 5 emails arrive in a particular 3-minute period. [4]

Answer:

(a) $X \sim Po(1.2)$

$(i)\ P(x \geq 2) \approx 0.337$

$(ii)\ Y \sim Po(3.6)$ $\lambda = 1.2 \times 3 = 3.6$ (for 3-minute period)

$P(Y = 5) \approx 0.138$

The manager records, for each of 10 separate one-minute intervals, whether fewer than 2 emails arrive. The intervals are assumed to be independent.

(b) (i) State an appropriate distribution to model the number of these intervals in which fewer than 2 emails arrive.

(ii) Hence, find the probability that this occurs in exactly 4 of the 10 intervals. [3]

Answer:

(b) Let W be the number of intervals in which fewer than 2 emails arrive.

(i) Using G.D.C. to find the probability in Poisson Distribution, we have

$$P(x < 2) = P(x \leq 1) = poissoncdf(1.2, 0, 1)$$

$$= 0.662627$$

Since there are fixed number of trials 10.

Only two outcomes, each interval has either fewer than 2 emails arrive, or it does not.

The probability of success (fewer than two emails) is constant, 0.662627

Each trial is independent as it is said in question.

Hence it is the Binomial Distribution:

$$\boxed{W \sim B(10, 0.662627)}$$

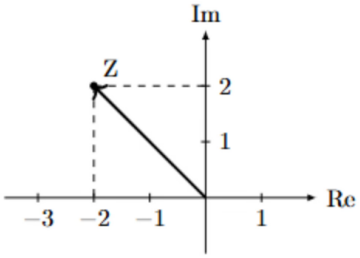
(ii) Using G.D.C.

$$P\left(w=4\right)=binomialpdf\left(10,0.662627,4\right)$$

$$=0.0597$$

[Maximum mark: 7]

The point Z on the following Argand diagram represents the complex number z.



(a) (i) Use the diagram to write down z in the form $a + bi$, where $a, b \in \mathbb{Z}$.

(ii) Hence, express z in the form $r\,e^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [3]

Consider the complex number $w = \sqrt{2}e^{i\left(-\frac{\pi}{3}\right)}$.

(b) Find $\text{Im}(zw)$ in the form $p\sin(q\pi)$, where $p \in \mathbb{Z}$ and $q \in \mathbb{Q}$. [4]

Answer:

(a)(i) $z = -2 + 2i$

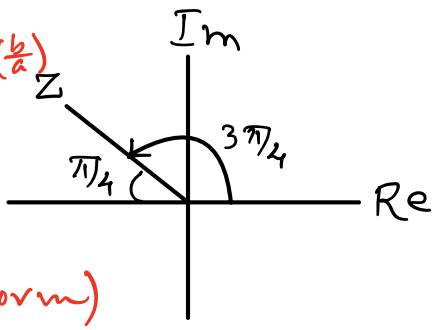
(ii) $|z| = \sqrt{4+4} = 2\sqrt{2} = r$

and

$$\theta = \tan^{-1}\left(\frac{-2}{2}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\therefore z = 2\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$$

(Euler form)



(b) $w = \sqrt{2}e^{i(-\pi/3)}$ and $z = 2\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$

$$zw = \left(2\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}\right)\left(\sqrt{2}e^{i\left(\frac{-\pi}{3}\right)}\right)$$

$$= 4e^{i\left(\frac{3\pi}{4} - \frac{\pi}{3}\right)}$$

$$= 4e^{i\left(\frac{5\pi}{12}\right)}$$

$$\therefore ZW = 4\cos\left(\frac{5\pi}{12}\right) + 4i\sin\left(\frac{5\pi}{12}\right)$$

(polar form)

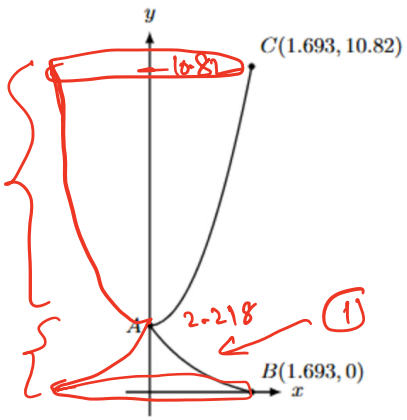
$$\therefore Im\,(zw) = 4\sin\left(\frac{5\pi}{12}\right)$$

Question 13

[Maximum mark: 7]

In this question, units are in centimetres.

The diagram below shows the function $y = e^{1-x} - 0.5$, and the function $y = 3x^2 + e - 0.5$, for $0 \leq x \leq 1+\ln 2$

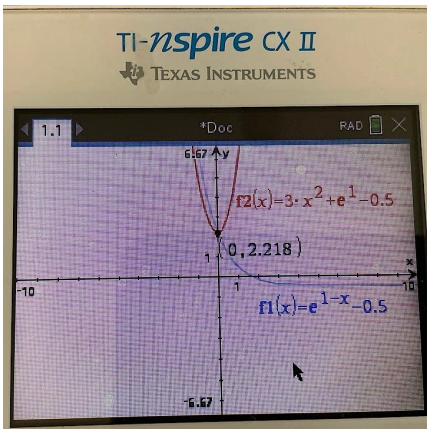


Point B has coordinates $(1.693, 0)$ and point C has coordinates $(1.693, 10.82)$, both correct to four significant figures. The two curves meet on the y-axis at point A.

(a) State the coordinates of point A, correct to four significant figures. [1]

Answer: Using G.D.C.

Coordinates of A $(0, 2.218)$



The sections of the curves shown are to be rotated 360° about the y-axis to form a glass.

(b) Using the four significant figure values of A, B and C, write an expression for this volume as the sum of two integrals. [4]

(c) Hence determine the volume of the glass. [2]

Answer:

Vol. about Y-axis $= \pi \int_a^b x^2 \, dy$

(b) From $y = e^{1-x} - 0.5$

$$y + 0.5 = e^{1-x}$$

$$1 - x = \ln(y + 0.5)$$

$$x = 1 - \ln(y + 0.5) \quad \text{——— (i)}$$

From $y = 0$ to $y = 2.218$

and from $y = 3x^2 + e - 0.5$

$$3x^2 = y - e + 0.5$$

$$x^2 = \frac{y - e + 0.5}{3}$$

from $y = 2.218$ to $y = 10.82$

$$\therefore Volume = \pi \int_0^{2.218} x^2 dy + \pi \int_{2.218}^{10.82} x^2 dy$$

$$\therefore Vol. = \pi \int_0^{2.218} (1 - \ln(y + 0.5))^2 dx + \pi \int_{2.218}^{10.82} \frac{(y - e + 0.5)}{3} dx$$

(c) Using G.D.C.

$$\therefore Vol. = 42.9 \ cm^3$$

Question 14

[Maximum mark: 9]

A New Zealand based internet service provider are constructing a new 5G network and aim to have 95% of their customers connected within a year.

The percentage of customers, P, connected to the new network after t weeks can be modelled by the following logistic equation:

$$P(t) = \frac{100}{1 + Ce^{-kt}}, \quad C, k > 0.$$

Initially, 25% of customers are connected to the network.

(a) Find the value of C. [1]

It is expected that 50% of customers will be connected after 12 weeks.

(b) (i) Find the value of k. Give your answer in the form $\frac{\ln a}{b}$, where $a, b \in \mathbb{Z}^+$.

(ii) Hence write $P(t)$ in the form $P(t) = \frac{100}{1 + C^{1-\frac{t}{n}}}$, where $n \in \mathbb{Z}^+$. [5]

(c) Determine whether the model indicates that the company will meet their aim. [3]

Answer:

(a) when t = 0, P(t) is 25%

$$\therefore \frac{100}{1+C} = 25 \Rightarrow 1 + C = 4 \Rightarrow C = 3$$

(b) when t = 12, P(t) = 50

(i) $$\therefore \frac{100}{1+3e^{-12k}} = 50 \Rightarrow 1 + 3e^{-12k} = 2 \Rightarrow 3e^{-12k} = 1$$

$$\Rightarrow e^{-12k} = \frac{1}{3} \Rightarrow -12k = \ln\left(\frac{1}{3}\right) = \overset{\checkmark}{-}\ln 3$$

$$\Rightarrow k = \frac{1}{12}\ln 3$$

(ii) $$P(t) = \frac{100}{1 + 3e^{-\frac{\ln 3}{12}t}} = \frac{100}{1 + 3\left(e^{\ln 3}\right)^{-\frac{t}{12}}} = \frac{100}{1 + 3\,(3)^{-\frac{t}{12}}}$$

$$\therefore P(t) = \frac{100}{1 + 3^{1-\frac{t}{12}}}$$

(c) $P(t) = \dfrac{100}{1 + 3^{1-\frac{t}{12}}}$

Company's target is to connect 95% of its customers within a year.

P(t) must reach 95 before t reaches a value of 52

Using G.D.C. to solve

$$\frac{100}{1 + 3^{1-\frac{t}{12}}} = 95$$

we get $t = 44.1617\ldots$

$\therefore \quad 44.1617\ldots < 52$

Yes, the model indicates that the company will meet their target.

Question 15

[Maximum mark: 6]

Consider two transformations, T_1 and T_2.

T_1 consists of a reflection in the line $y = \frac{1}{\sqrt{3}}x$ followed by a rotation of 90° anticlockwise about the origin.

T_2 consists of an enlargement by a scale factor of $\frac{1}{2}$ from the origin, followed by a translation by the vector $\begin{pmatrix} 1+\sqrt{3} \\ -1 \end{pmatrix}$.

It is given that the image of the point (p, q) under T_1 is the same as its image under T_2.

Find the value of p and the value of q. [6]

Answer:

Reflection in the line y = (tanθ) x , can be expressed as $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

AHL 3.9 | Transformation matrices | $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$, reflection in the line $y = (\tan\theta)x$

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, anticlockwise/counter-clockwise rotation of angle θ about the origin ($\theta > 0$)

We have $y = \frac{1}{\sqrt{3}}x \Rightarrow \tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$

Therefore reflection matrix is $\begin{pmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{pmatrix}$

A rotation in anticlockwise direction about origin is given by $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

Hence for 90° anticlockwise rotation the matrix is $\begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{pmatrix}$

Since T_1 first reflects in the line at 30° to positive x-axis and then rotates by 90° anticlockwise

$$\therefore T_1\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90 & \cos 90^\circ \end{pmatrix}\begin{pmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix}$$

Now $T_1\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} -\sqrt{3}/2 & 1/2 \\ 1/2 & \sqrt{3}/2 \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} \rightarrow$

T_2 consists of an enlargement by a scale factor of $\frac{1}{2}$ from the origin, followed by a translation by the vector $\begin{pmatrix} 1+\sqrt{3} \\ -1 \end{pmatrix}$.

$$\therefore T_2\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} + \begin{pmatrix} 1+\sqrt{3} \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}p \\ \frac{1}{2}q \end{pmatrix} + \begin{pmatrix} 1+\sqrt{3} \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}p+1+\sqrt{3} \\ \frac{1}{2}q-1 \end{pmatrix}$$

Given that

It is given that the image of the point (p, q) under T_1 is the same as its image under T_2.

$$\therefore T_1\begin{pmatrix} p \\ q \end{pmatrix} = T_2\begin{pmatrix} p \\ q \end{pmatrix}$$

Using G.D.C.

$$T_1\begin{pmatrix}p\\q\end{pmatrix}=\begin{pmatrix}-0.866025 & 0.5\\0.5 & 0.866025\end{pmatrix}\begin{pmatrix}p\\q\end{pmatrix}$$

$$\begin{pmatrix}-0.866025 & 0.5\\0.5 & 0.866025\end{pmatrix}\begin{pmatrix}p\\q\end{pmatrix}=\begin{pmatrix}0.5 & 0\\0 & 0.5\end{pmatrix}\begin{pmatrix}p\\q\end{pmatrix}+\begin{pmatrix}1+\sqrt{3}\\-1\end{pmatrix}$$

Solving two linear equations

$$-0.866025p+0.5q=0.5p+2.732050808$$

$(-\frac{\sqrt{3}}{2}=-0.866025\ldots)$

&

$$0.5p+0.866025q=0.5q-1$$

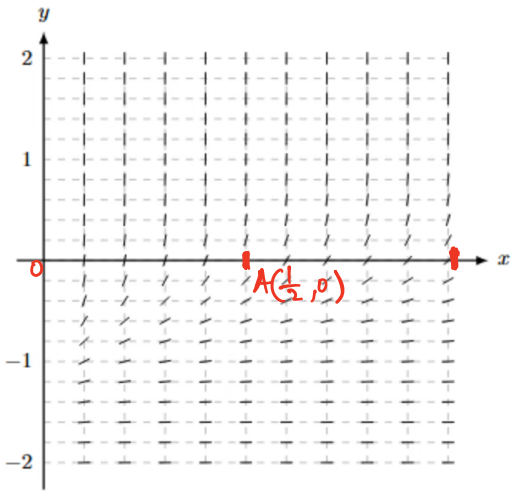
Using G.D.C.

$$p=-2, q=0$$

Question 16

[Maximum mark: 8]

The slope field for the differential equation $\frac{dy}{dx} = \frac{e^{3y}}{x}$, where $0 < x < 1$, is shown in the following graph.



Point A has coordinates $\left(\frac{1}{2}, 0\right)$.

(a) Find the gradient of the solution curve at A. [2]

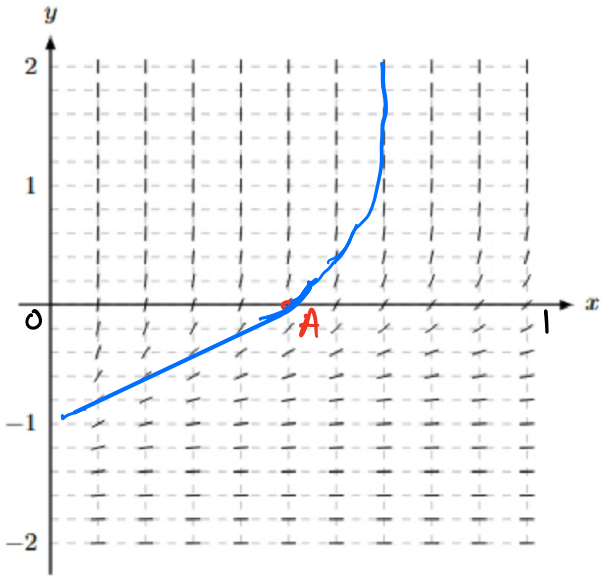
Answer:

(a) gradient at A(0.5,0) is $\frac{dy}{dx} = \frac{e^{3y}}{x}$;

$$\frac{dy}{dx} = \frac{e^0}{\left(\frac{1}{2}\right)} = 2$$

One solution to $\frac{dy}{dx} = \frac{e^{3y}}{x}$ passes through A.

(b) Sketch this solution on the slope field.



The general solution curve to the differential equation is $y = -\frac{1}{3}\ln\left(C - 3\ln(x)\right)$, where $C \in \mathbb{R}$ and $x > 0$

A particular solution passes through the point $(e, 0)$ and has vertical asymptotes of $x = 0$ and $x = e^k$, where $k \in \mathbb{Q}$.

(c) Find k. [4]

Answer:

(c) Substituting $x = e, y = 0$, we get

$$0 = -\frac{1}{3}\ln(c - 3\ln e)$$

$$\ln(C - 3) = 0$$

$$\therefore C - 3 = 1$$

$$C = 4$$

Therefore particular solution is

$$y = -\frac{1}{3}\ln(4 - 3\ln(x))$$

In logarithm

$$4 - 3\ln x > 0$$

Now, vertical asymptote occurs at x where the function is undefined.

$$\therefore 4 - 3\ln x = 0$$

$$\Rightarrow \ln x = \frac{4}{3}$$

$$\Rightarrow x = e^{\frac{4}{3}}$$

$$\therefore k = \frac{4}{3}$$

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